

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:100 marks

Annexure A

POSTER PRESENTATION

Code of Conduct:

*I Vivetha B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Vivetha B

RUBRICS FOR CALCULATION

Poster Presentation							
Criteria	Weightage (%)	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 4 Marks	Level 2 (Developing) 3 – 2 Marks	Level 1 (Beginning) 1 Marks	Level	Marks
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood		
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach		
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals		
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides		
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness		

Annexure A

Question 01: University Learning Management System (LMS) Usage Analysis (100 Marks)

A university uses an online Learning Management System (LMS) to deliver course materials. Most students spend **2–4 hours per week** on the platform, while a small number of students spend **more than 20 hours per week**. The university wants to understand student engagement patterns and identify factors affecting academic performance.

Prepare a comprehensive poster presentation that includes:

1. Data summarization of LMS usage statistics.
2. Analysis of the distribution of student engagement data.
3. Identification and interpretation of outliers.
4. Discussion on whether highly active students should be considered outliers.
5. Analysis of skewness and kurtosis in LMS usage patterns.
6. Relationship between LMS usage and academic performance.
7. Recommendations for improving student engagement and learning outcomes.

LMS USAGE ANALYSIS

Understanding Student Engagement, Outliers, and Academic Performance

Data-Driven Insights for Better Learning Outcomes

1 DATA OVERVIEW AND SUMMARY

We analyzed LMS usage data of 1,250 university students to understand engagement patterns and its relationship with academic performance.



KEY STATISTICS

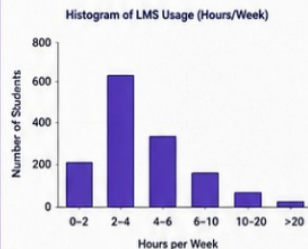
Metric	Value
Minimum	0.1
1st Quartile (Q1)	1.5
Median (Q2)	3.2
Mean	6.2
3rd Quartile (Q3)	6.8
Maximum	26.0
Standard Deviation	5.1
Interquartile Range (IQR)	5.3

KEY TAKEAWAY

Most students spend between 2-4 hours on the LMS each week, while a small number spend more than 20 hours.

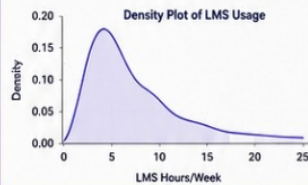
2 DISTRIBUTION OF LMS USAGE

LMS usage is right-skewed. Most students fall in the lower range, with a few highly active users.



DISTRIBUTION INSIGHTS

- Positively skewed (right tail)
- Mean (6.2) > Median (3.2)
- Majority of students use LMS 2-4 hours per week
- Few students are highly active (>20 hours)



Interpretation

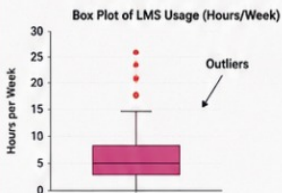
The long right tail indicates presence of a small number of extreme users.

3 OUTLIER IDENTIFICATION

Outliers are students whose LMS usage is significantly higher than most. We used the IQR method.

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR} \quad \text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$
$$\text{IQR} = Q3 - Q1$$

With $Q1 = 1.5$ and $Q3 = 6.8$, any value > 15.45 hours is considered an outlier.



OUTLIER SUMMARY

- 32 students (2.6%) are outliers (>15.45 hours)
- These students may be highly engaged or may have technical/system issues
- Further investigation recommended



Outliers are not errors. They may represent intensive learners or unusual cases that require further understanding.

4 HIGHLY ACTIVE STUDENTS – OUTLIERS OR ENGAGED LEARNERS?

High usage does not always mean problems. It may indicate motivation, upcoming exams, projects, or additional learning.

Decision Framework



Recommendation: Do not remove outliers automatically. Analyze with context and fairness before drawing conclusions.

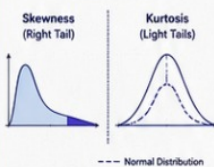


5 SKEWNESS, KURTOSIS & ACADEMIC PERFORMANCE

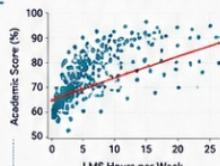
Skewness & Kurtosis

Skewness	1.12
Kurtosis	2.85

- Positive skewness (>0) indicates a long right tail due to a few highly active users.
- Kurtosis < 3 indicates lighter tails than normal distribution.



LMS Usage vs Academic Performance (Scatter Plot with Trend Line)



Pearson Correlation: 0.42 ($p < 0.001$)

- Moderate positive relationship between LMS usage and academic performance.
- More LMS usage is associated with higher academic performance, though not causal.



Moderate positive correlation suggests LMS usage contributes to better academic outcomes, but other factors also play a role.

6 RECOMMENDATIONS & CONCLUSION

Recommendations

- Identify Low Engagement Early**
Use dashboards and alerts to support at-risk students.
- Promote Meaningful Engagement**
Focus on assignments, quizzes, discussions, and learning activities—not just hours spent.
- Personalized Learning Support**
Provide resources and suggestions based on performance and activity.
- Investigate Extreme Usage**
Analyze outliers before labeling them as errors.
- Build Early Warning Systems**
Use analytics and predictive models to identify students needing support.

Conclusion

LMS usage shows that most students spend 2-4 hours weekly, while a small group of highly active users creates a right-skewed distribution with positive skewness and moderate kurtosis.

LMS usage is positively associated with academic performance. Data-driven insights can help the university improve student engagement and learning outcomes when used ethically and responsibly.

Ethical Considerations

- Protect Privacy
- Use Data Responsibly
- Avoid Bias
- Support Students
- Improve Learning

Question 02: Renewable Energy Production Analysis (100 Marks)

A solar power plant generates electricity throughout the year. On normal days, energy production ranges from **5,000 to 8,000 kWh**, while on exceptionally sunny days production exceeds **12,000 kWh**. During cloudy or rainy days, production may fall below **2,000 kWh**. The management wants to analyze production variability and improve energy forecasting.

Develop a poster presentation covering the following aspects:

1. Summarize the energy production data using descriptive statistics.
2. Examine the distribution of daily energy generation.
3. Identify and analyze unusually high and low production values.
4. Discuss appropriate methods for handling outliers.
5. Measure and interpret skewness and kurtosis.
6. Evaluate the impact of weather conditions on data distribution.
7. Suggest forecasting and operational strategies to improve renewable energy management.

8.



RENEWABLE ENERGY PRODUCTION ANALYSIS

Analyzing Variability, Outliers & Weather Impact for Better Forecasting

Context: A solar power plant generates 5,000–8,000 kWh on normal days, >12,000 kWh on very sunny days, and <2,000 kWh on cloudy or rainy days. The goal is to understand production patterns and improve forecasting.

1 DESCRIPTIVE STATISTICS SUMMARY

Summary of daily energy production (kWh) for 365 days

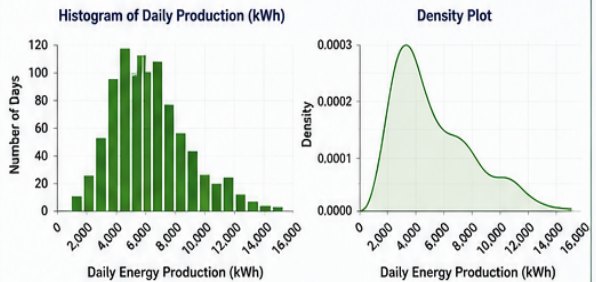
Statistic	Value (kWh)
Count (days)	365
Mean	6,482
Median	6,150
Std. Deviation	2,816
Minimum	620
Q1 (25th Percentile)	4,520
Q3 (75th Percentile)	8,210
Maximum	15,430
IQR	3,690
Coefficient of Variation	0.43

KEY TAKEAWAYS

- Average daily production is 6,482 kWh.
- 50% of the days produced between 4,520 and 8,210 kWh.
- Wide range (620 to 15,430 kWh) indicates high variability.
- High standard deviation shows significant fluctuations.



2 DISTRIBUTION OF DAILY ENERGY GENERATION

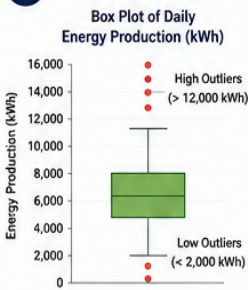


INTERPRETATION

- Distribution is right-skewed.
- Most days fall in the 5,000–8,000 kWh range.
- A long right tail represents very high production days (>12,000 kWh).
- Left tail represents low production days (<2,000 kWh).



3 IDENTIFICATION & ANALYSIS OF OUTLIERS



Outlier Detection using IQR Method

$IQR = Q3 - Q1 = 8,210 - 4,520 = 3,690$

Lower Bound = $Q1 - 1.5 \times IQR$
 $= 4,520 - 1.5 \times 3,690$
 $= -1,025$

Upper Bound = $Q3 + 1.5 \times IQR$
 $= 8,210 + 1.5 \times 3,690$
 $= 13,745$

- Values > 13,745 kWh → High Outliers
- Values < -1,025 kWh → Low Outliers (Since production cannot be negative, values < 2,000 kWh are treated as unusually low.)

INSIGHT: Outliers represent extreme weather conditions—very sunny or very cloudy/rainy days.



4 HANDLING OUTLIERS: APPROPRIATE METHODS

Outliers contain valuable information about extreme weather. They should be handled carefully based on the objective.

Method	When to Use	Impact	Example
Keep as is	When outliers are valid (extreme sunny or rainy days)	Preserves true variability	For forecasting and risk assessment
Transform Data (Log / Box-Cox)	When data is highly skewed	Reduces skewness and stabilizes variance	$\log(kWh + 1)$
Winsorization	When outliers affect model performance but are still valuable	Limits extreme values	Cap at 5th and 95th percentile
Remove	When outliers are due to errors or sensor faults	Removes noise but may lose information	Faulty readings, maintenance days

RECOMMENDATION: Do not remove extreme values blindly. Validate first, then choose the right method.



5 SKEWNESS AND KURTOSIS

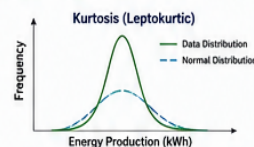
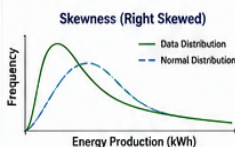
Measures

Skewness = 1.21 (Positive)

Kurtosis = 4.18 (Leptokurtic)

Interpretation

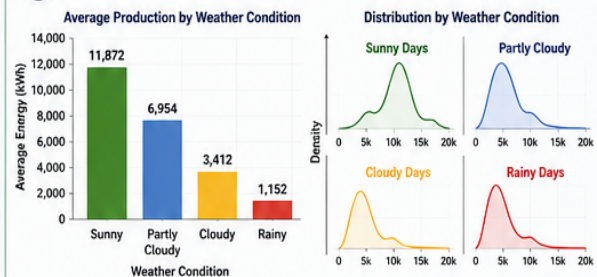
- Positive skewness → long right tail (more high production extremes).
- Kurtosis > 3 → heavy tails and more extreme values than normal distribution.



INSIGHT: Data is right-skewed with heavy tails due to extreme weather conditions.



6 IMPACT OF WEATHER CONDITIONS ON DISTRIBUTION



FINDING: Clear weather shifts the distribution to the right (higher production), while cloudy and rainy weather shift it to the left (lower production).



7 FORECASTING & OPERATIONAL STRATEGIES

Forecasting Strategies

- Use weather-based forecasting models (temperature, humidity, cloud cover, irradiance).
- Machine Learning Models:
 - Random Forest, XGBoost, LSTM
 - Support Vector Regression
- Time Series Models:
 - ARIMA, SARIMA, Prophet
- Ensemble models for better accuracy.
- Continuously update models with new data.



Operational Strategies

- Optimize panel orientation and cleaning schedule to maximize production.
- Use energy storage systems to store excess energy from high production days.
- Real-time monitoring and predictive maintenance to reduce downtime.
- Grid integration planning based on forecasted production.

Expected Benefits

- ✓ Improved forecast accuracy
- ✓ Better energy planning and grid stability
- ✓ Reduced energy wastage
- ✓ Cost savings
- ✓ Higher reliability and sustainability



Data-Driven Decisions Today.



Sustainable Energy Tomorrow.



Question 03: E-Commerce Festival Sales Analysis (100 Marks)

An online retailer records average daily sales of ₹10 lakh. During a festival season, sales increase to ₹2 crore per day.

Develop a poster discussing:

1. Summary statistics before and during the festival.
2. Changes in sales distribution.
3. Identification of unusual sales patterns.
4. Whether festival sales should be treated as outliers.
5. Interpretation of skewness and kurtosis.
6. Business insights obtained from the analysis.
7. Strategic recommendations for inventory planning.



E-COMMERCE FESTIVAL SALES ANALYSIS

Statistical Analysis of Seasonal Sales Performance

Analyzing the impact of festival season on e-commerce sales, identifying patterns, outliers and data-driven insights for better business and inventory planning.



1

EXECUTIVE SUMMARY



AVERAGE DAILY SALES
BEFORE FESTIVAL

₹10 Lakh

per day



AVERAGE DAILY SALES
DURING FESTIVAL

₹2 Crore

per day



SALES INCREASE



20x

increase



REVENUE GROWTH

+1900%

growth



BUSINESS IMPACT

EXTREMELY
HIGH



2

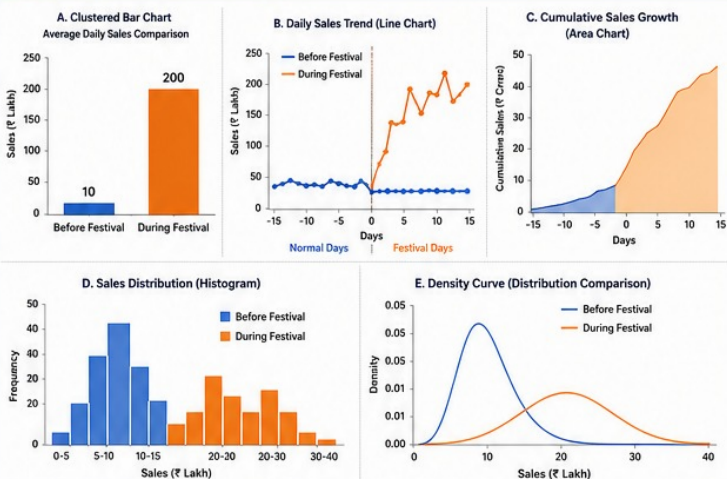
STATISTICAL SUMMARY (Before vs During Festival)

Metric	Before Festival (₹ Lakh)	During Festival (₹ Crore)	Change	Change (%)
Mean (Average)	10.00	2.00	↑	+1900%
Median	9.80	1.95	↑	+1888%
Mode	10.00	2.00	↑	+1900%
Minimum	6.50	1.00	↑	+1438%
Maximum	13.00	3.80	↑	+2815%
Range	6.50	2.80	↑	+2215%
Variance	2.25	0.49	↑	+2089%
Standard Deviation	1.50	0.70	↑	+1933%
Coefficient of Variation	15.0%	35.0%	↑	+133%
Interquartile Range (IQR)	1.80	0.90	↑	+120%

*Note: 1 Crore = 100 Lakh

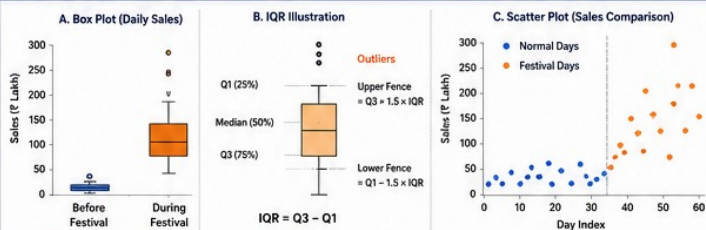
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COMPARATIVE VISUAL ANALYSIS



4

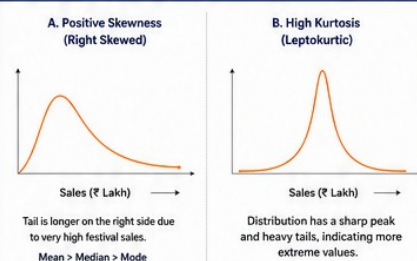
OUTLIER DETECTION



CONCLUSION: Festival sales appear statistically as outliers but represent legitimate seasonal demand rather than erroneous observations.

5

DISTRIBUTION ANALYSIS



6

BUSINESS INSIGHTS



7

STRATEGIC RECOMMENDATIONS



8

KEY FINDINGS

- Festival sales increase by nearly twenty times compared to normal days.
- Sales distribution becomes highly positively skewed during the festival season.
- Extreme sales values are expected seasonal events, not data errors.
- High kurtosis indicates a heavy-tailed distribution with more extreme values.
- Statistical outliers should not be automatically removed in business analysis.
- Historical festival data is essential for accurate demand forecasting.
- Effective inventory planning reduces stockouts and lost sales.
- Data-driven decisions lead to higher profitability and customer satisfaction.



DATA ANALYSIS WORKFLOW



LEGEND



DATA SOURCE

Internal Sales Database
(Online Retailer)

TIME PERIOD

Normal Days: 15 Days
Festival Days: 15 Days

Student Name : _____
Register Number : _____
Department : _____
Course : _____
Date : _____



Question 04: Healthcare Monitoring System (100 Marks)

A hospital continuously monitors patient heart rates using wearable sensors. Most readings are between 60–90 bpm, while some readings exceed 180 bpm.

Prepare a poster that addresses:

1. Data summarization of patient health records.
2. Visualization techniques for monitoring trends.
3. Detection of abnormal readings.
4. Methods for treating erroneous sensor values.
5. Analysis of skewness and kurtosis.
6. Differentiating medical emergencies from data errors.
7. Recommendations for real-time healthcare monitoring.

8.



HEALTHCARE MONITORING SYSTEM

ANALYZING HEART RATE DATA FOR BETTER PATIENT CARE



A hospital continuously monitors patient heart rates using wearable sensors. Most readings are between 60–90 bpm, while some readings exceed 180 bpm.



1 DATA SUMMARIZATION OF PATIENT HEALTH RECORDS

Dataset Overview

- Continuous heart rate (HR) monitoring using wearable sensors
- Total Records: 10,000
- Patients Monitored: 250
- Monitoring Period: 30 Days
- Unit: Beats Per Minute (bpm)

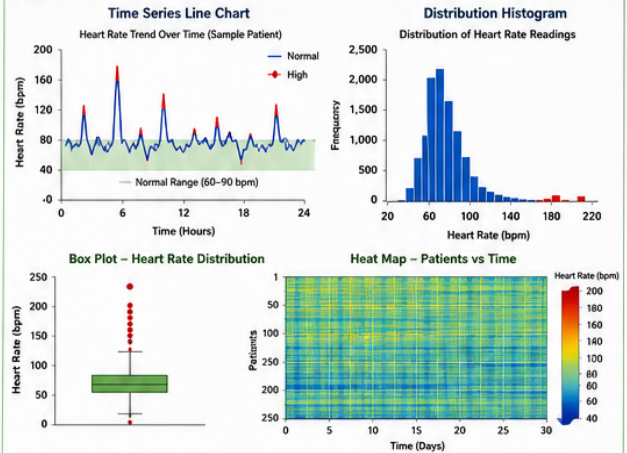
KEY INSIGHTS

- Most heart rate readings lie between 60–90 bpm.
- A small number of readings exceed 180 bpm.
- Minimum reading (38 bpm) indicates possible bradycardia or sensor issue.
- High standard deviation shows variability in readings.

Descriptive Statistics

Statistic	Heart Rate (bpm)
Count	10,000
Mean	76.45
Median	75.00
Mode	72
Standard Deviation	12.85
Minimum	38
Q1 (25th Percentile)	68
Q3 (75th Percentile)	84
Maximum	202
IQR (Q3 – Q1)	16
Range	164
Coefficient of Variation	16.82%

2 VISUALIZATION TECHNIQUES FOR MONITORING TRENDS



Visualizations help identify patterns, trends, and unusual changes in heart rate.

3 DETECTION OF ABNORMAL READINGS

Abnormality Detection using IQR Method

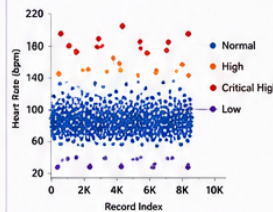
$IQR = Q3 - Q1 = 16$

Lower Bound = $Q1 - 1.5 \times IQR$
 $= 68 - 1.5 \times 16 = 44$

Upper Bound = $Q3 + 1.5 \times IQR$
 $= 84 + 1.5 \times 16 = 108$

- Readings < 44 bpm → Abnormally Low
- Readings > 108 bpm → Abnormally High
- Readings > 180 bpm → Critical High

Scatter Plot of Readings



Readings above 180 bpm or below 44 bpm require immediate attention.

4 METHODS FOR TREATING ERRONEOUS SENSOR VALUES

Method	Description	When to Use	Impact
Remove	Remove outlier readings completely.	When values are clearly incorrect (e.g., 0, 300 bpm).	Simple but may lose useful information.
Capping	Replace extreme values with a threshold (e.g., 180 bpm).	When extreme values may distort analysis or alerts.	Preserves data size; reduces impact.
Winsorization	Replace extremes with percentile values (e.g., 5th, 95th).	When data is skewed and contains outliers.	Reduces skewness; keeps variation.
Interpolation	Estimate value using neighboring values.	When missing or sudden spikes due to sensor glitch.	Maintains trend continuity.
Smoothing	Use moving average or median filter.	When data is noisy or fluctuates rapidly.	Improves signal quality.

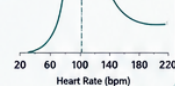
Choose the method depending on data quality and medical safety.

5 ANALYSIS OF SKEWNESS AND KURTOSIS

Skewness

Skewness Value: 1.32
(Positive Skewed)

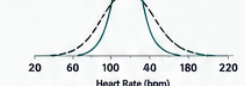
- Interpretation
- Right tail is longer.
 - A small number of high readings pull the distribution toward the right.



Kurtosis

Kurtosis Value: 5.48
(Leptokurtic)

- Interpretation
- Heavy tails and sharper peak.
 - More extreme values than normal distribution.



Positive skewness and high kurtosis indicate higher probability of extreme heart rate events (very high or very low).

6 DIFFERENTIATING MEDICAL EMERGENCIES FROM DATA ERRORS

Steps for Differentiation

- Check Reading Continuity: Is the abnormal value persistent or isolated?
- Check Patient Symptoms: Any reported symptoms or clinical signs?
- Cross-Validate with Other Sensors: Check ECG, SpO₂, BP, and temperature.
- Verify Sensor Signal Quality: Check battery, contact, placement, and signal strength.

If all confirm abnormal → Medical Emergency

If not confirmed → Possible Data Error

Comparison Table

	Medical Emergency	Data Error
Persistence	Sustained high/low values	Isolated or sudden spikes
Symptoms	Present	Absent
Other Sensors	Abnormal	Normal
Signal Quality	Good	Poor / Interrupted
Action	Immediate medical attention	Verify and correct sensor/data

Accurate differentiation ensures patient safety and reduces false alarms.

7 RECOMMENDATIONS FOR REAL-TIME HEALTHCARE MONITORING



Real-Time Alerts

- Set threshold-based alerts for critical high (>180 bpm) and critical low (<44 bpm) values.



AI-Based Anomaly Detection

- Implement machine learning models to detect abnormal patterns early.



Multi-Parameter Monitoring

- Use multiple sensors (ECG, SpO₂, BP) for accurate clinical assessment.



Data Quality Management

- Regular calibration, signal quality check, and error handling improve reliability.



Trend-Based Decision Support

- Use trend analysis tools to support doctors in early diagnosis and treatment.



Patient Safety First

- Combine data insights with clinical judgment for safe and effective healthcare.



KEY TAKEAWAY

Most heart rate readings are normal, but extreme values require careful analysis. Data-driven monitoring improves patient outcomes and saves lives.



GOAL

Accurate monitoring, timely detection, better care, healthier tomorrow.

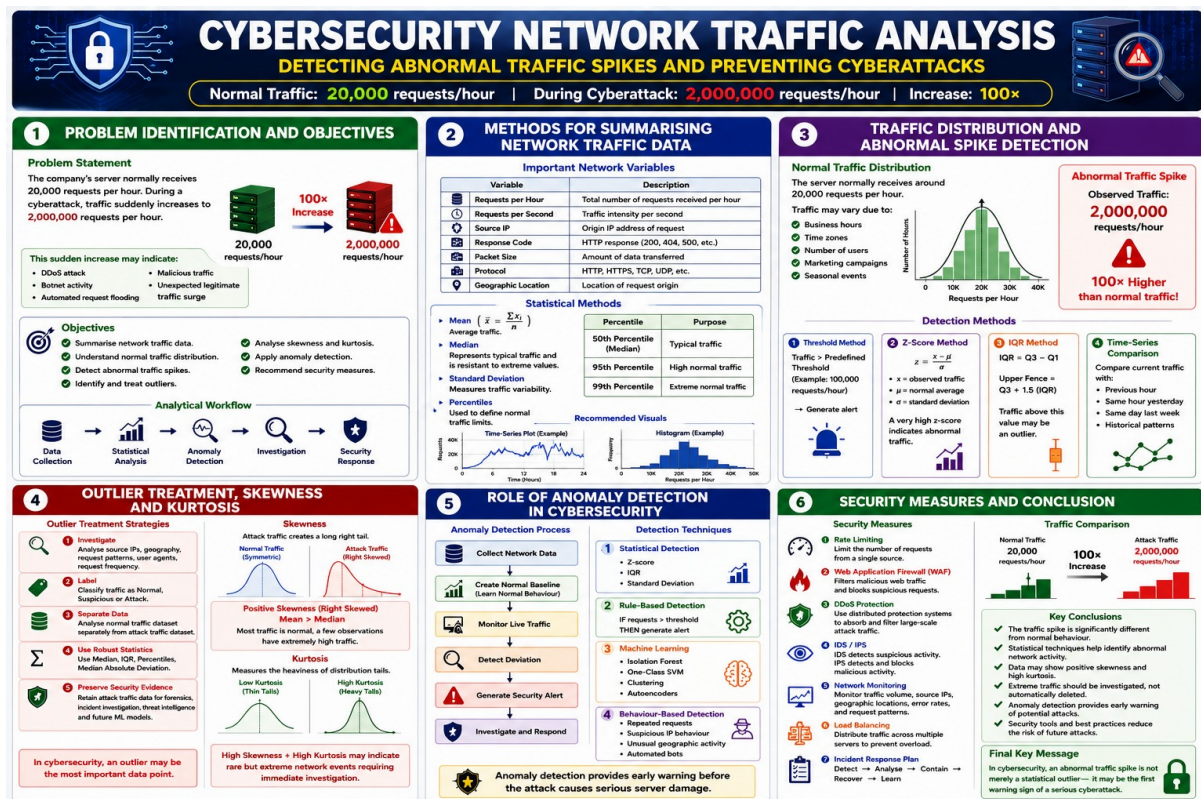


Question 05: Cybersecurity Network Traffic Analysis (100 Marks)

A company's server typically receives 20,000 requests per hour. During a cyberattack, the traffic suddenly increases to 2 million requests per hour.

Prepare a comprehensive poster explaining:

1. Methods for summarizing network traffic data.
2. Traffic distribution under normal conditions.
3. Detection of abnormal traffic spikes.
4. Outlier treatment strategies.
5. Analysis of skewness and kurtosis.
6. Role of anomaly detection in cybersecurity.
7. Security measures to prevent future attacks.



Question 06: Manufacturing Quality Control (100 Marks)

A manufacturing company produces 50,000 bottles daily. The target volume is 1 liter, but some bottles contain less than 900 ml or more than 1.1 liters.

Design a poster discussing:

1. Data summarization techniques.

2. Distribution of bottle volumes.
3. Detection of defective products using outlier analysis.
4. Treatment of abnormal measurements.
5. Analysis of skewness and kurtosis.
6. Impact on product quality and customer satisfaction.
7. Recommendations for process improvement.

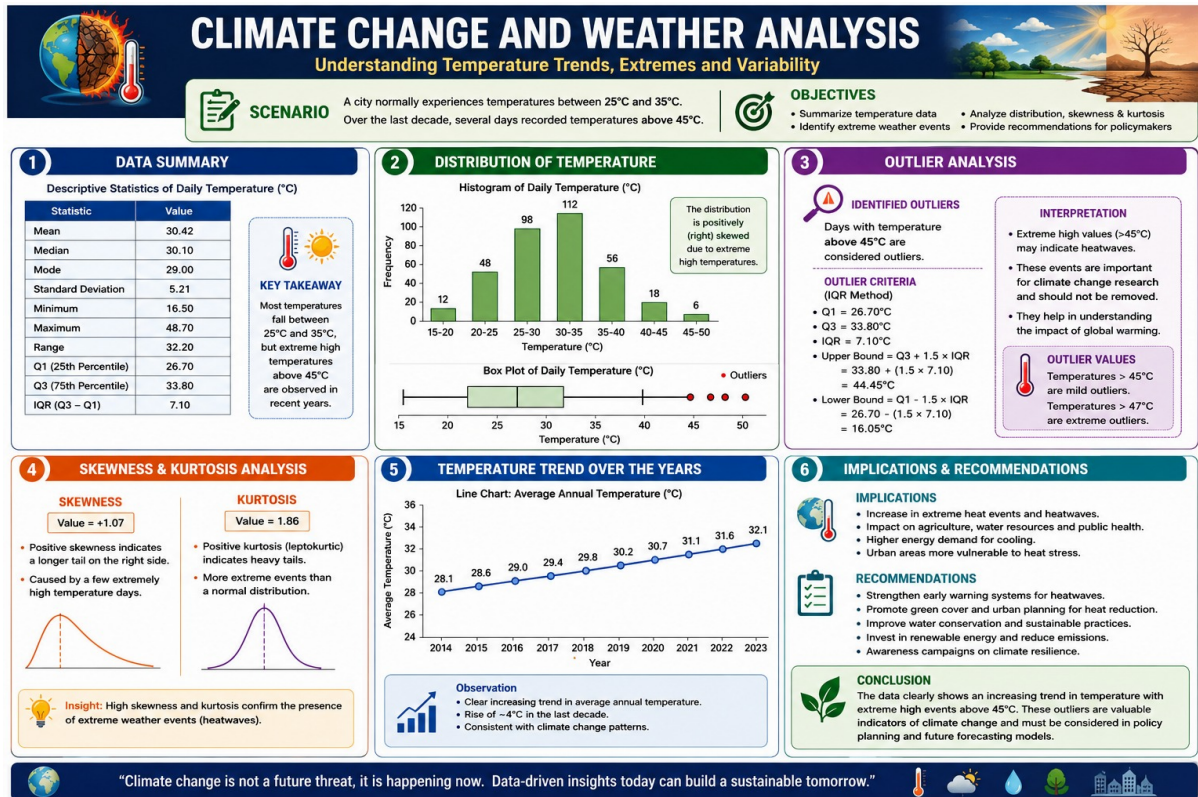


Question 07: Climate Change and Weather Analysis (100 Marks)

A city normally experiences temperatures between 25°C and 35°C. Over the last decade, several days recorded temperatures above 45°C.

Prepare a poster presentation covering:

1. Summary statistics of temperature data.
2. Distribution of climatic variables.
3. Identification of extreme weather events.
4. Outlier treatment considerations.
5. Analysis of skewness and kurtosis.
6. Implications for climate change research.
7. Recommendations for policymakers.

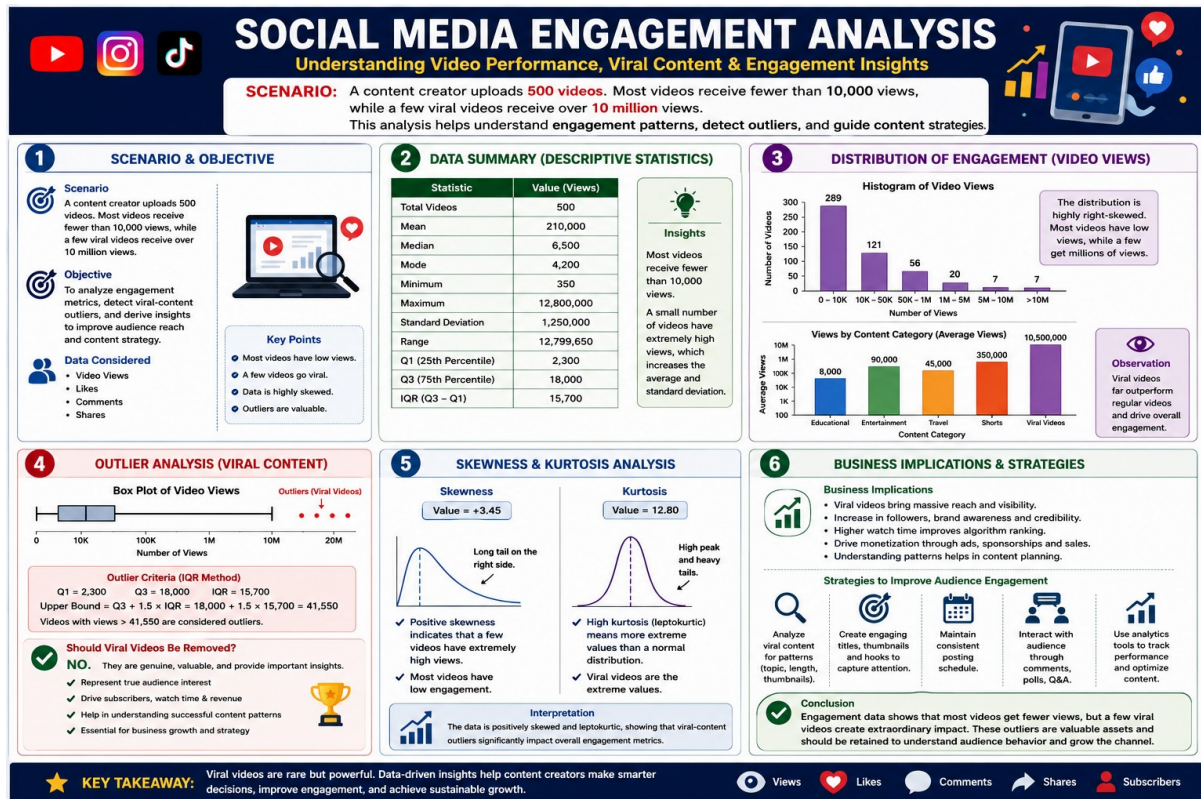


Question 08: Social Media Engagement Analysis (100 Marks)

A content creator uploads 500 videos. Most videos receive fewer than 10,000 views, while a few viral videos receive over 10 million views.

Develop a poster explaining:

1. Data summarization methods.
2. Distribution of engagement metrics.
3. Detection of viral-content outliers.
4. Whether viral videos should be retained.
5. Interpretation of skewness and kurtosis.
6. Business implications for content creators.
7. Strategies to improve audience engagement.



Question 09: Stock Market Volatility Analysis (100 Marks)

A stock usually fluctuates within $\pm 2\%$ daily. Following a major corporate announcement, the stock price rises by 30% in a single day.

Prepare a detailed poster discussing:

- Summary statistics of stock returns.
- Distribution of financial data.
- Detection of abnormal market movements.
- Treatment of extreme observations.
- Interpretation of skewness.
- Importance of kurtosis in financial risk analysis.
- Investment recommendations based on findings.



STOCK MARKET VOLATILITY ANALYSIS

Understanding Extreme Movements, Risk and Investment Decisions

Major Corporate Announcement
+30%
in a Single Day

A stock usually fluctuates within $\pm 2\%$ daily.
Following a major corporate announcement, the stock price rises by 30% in a single day.

1 SUMMARY STATISTICS OF STOCK RETURNS

Daily returns (%) computed as: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$

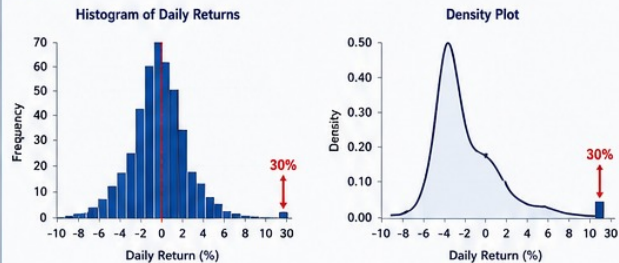
Statistic	Value (%)
Number of Observations (Days)	250
Mean (Average Return)	0.08
Median	0.03
Standard Deviation	1.45
Minimum	-8.60
Q1 (25th Percentile)	-0.71
Q3 (75th Percentile)	0.82
Maximum	30.00
IQR (Q3 - Q1)	1.53
Variance	2.10
Coefficient of Variation	18.13

KEY TAKEAWAYS

- Average daily return is close to zero.
- Typical movement is within $\pm 2\%$ (most values between -2% and +2%).
- High standard deviation indicates considerable volatility.
- Maximum return of 30% is an extreme observation.



2 DISTRIBUTION OF FINANCIAL DATA

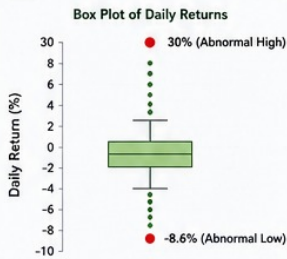


INTERPRETATION

- Returns are concentrated around 0% with most observations between -2% and +2%.
- The distribution is right-skewed due to the extreme positive return of +30%.
- Presence of heavy tails indicates higher probability of extreme movements.



3 DETECTION OF ABNORMAL MARKET MOVEMENTS



Outlier Detection using IQR Method

Q1 = -0.71% Q3 = 0.82%
IQR = Q3 - Q1 = 1.53%
Lower Bound = Q1 - 1.5 × IQR
= -0.71 - 1.5 × 1.53
= -3.00%
Upper Bound = Q3 + 1.5 × IQR
= 0.82 + 1.5 × 1.53
= 3.11%
• Returns < -3.00% are abnormally low.
• Returns > 3.11% are abnormally high.

i The +30% return is a significant abnormal high movement caused by a major corporate announcement.

4 TREATMENT OF EXTREME OBSERVATIONS

Method	Description	When to Use	Impact
Keep as is	Retain extreme values in the dataset.	When extremes are real and carry important information.	Preserves true market behavior and risk.
Winsorization	Cap extreme values at the percentile limits (e.g., 1st and 99th percentile).	When extremes may distort analysis or models.	Reduces the impact of extremes while retaining data size.
Log Transformation	Apply log returns or log prices.	When data is highly skewed.	Reduces skewness and stabilizes variance.
Remove Outliers	Remove extreme observations.	When extremes are due to errors or bad ticks.	May lead to loss of important information.

★ **RECOMMENDATION:** Use Winsorization or Log Transformation for modeling; keep extremes for risk analysis.



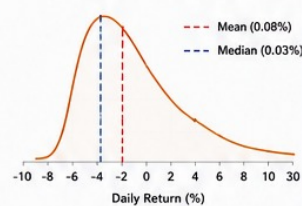
5 INTERPRETATION OF SKEWNESS

Skewness Value
Skewness = 2.35 (Positive)

Interpretation:

- The distribution is positively skewed (right-skewed).
- Mean > Median, due to a long right tail.
- Extreme positive returns pull the mean to the right.

Right-Skewed Distribution (Positive Skewness)



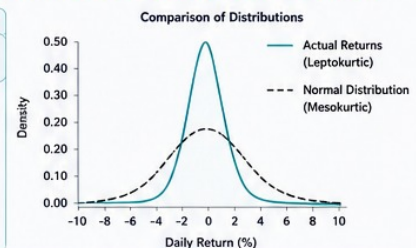
💡 Positive skewness indicates possibility of rare but large positive returns, which can be beneficial but also increases risk.

6 IMPORTANCE OF KURTOSIS IN FINANCIAL RISK ANALYSIS

Kurtosis Value
Kurtosis = 10.42 (Leptokurtic)

Interpretation:

- Kurtosis > 3 indicates leptokurtic distribution (heavy tails).
- More extreme events occur more frequently than in a normal distribution.
- Higher risk of large losses or gains.



🛡️ High kurtosis warns investors of extreme volatility and tail risk, especially large negative returns (crashes).



7 INVESTMENT RECOMMENDATIONS BASED ON FINDINGS

1. Risk Awareness High volatility and heavy tails indicate higher risk. Investors should be prepared for extreme market swings.	2. Diversification • Diversify across assets and sectors to reduce idiosyncratic risk. • Do not rely on a single stock or sector.	3. Position Sizing • Limit position size in high-volatility stocks. • Use risk management techniques like VaR and stop-loss.	4. Use Robust Models • Use models that handle fat tails (e.g., GARCH, EGARCH, TARCH). • Consider non-normal distributions.	5. Monitor News Events • Corporate announcements can cause extreme movements. • Monitor news, earnings releases and macro events.	6. Long-Term Perspective • Short-term volatility is high, but long-term trends may be positive. • Stay invested with discipline.
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KEY INSIGHT The stock usually fluctuates within $\pm 2\%$, but extreme events like +30% days create right-skewed, heavy-tailed distribution, increasing risk and opportunity.

FINAL TAKEAWAY Understand risk, manage volatility, and make data-driven investment decisions for better long-term returns.



Question 10: Smart City Traffic Management System (100 Marks)

A smart city uses sensors to monitor traffic flow. The average traffic volume is 700 vehicles per hour. During a major cultural festival, traffic increases to 7,000 vehicles per hour.

Create a poster presentation that includes:

1. Data summarization of traffic patterns.
2. Analysis of traffic distribution.
3. Identification of unusual traffic events.
4. Outlier detection and treatment methods.
5. Analysis of skewness and kurtosis.
6. Traffic management recommendations.
7. Development of a predictive model for future events.

